Northwestern

Wildcat Beamer Theme

Aaron Wolf (Northwestern University)

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Introduction

The Wildcat theme is a Beamer theme for Northwestern University, but which can be modified easily with different colors, fonts, and even background patterns.

The theme is inspired by the Metropolis theme by Matthias Vogelgesang. It incorporates the Northwestern University facet design pattern, but otherwise has a clean, simple look, and relatively few bells and whistles. It is licensed under the GNU GENERAL PUBLIC LICENSE.

Background Patterns

The theme ships with 12 built-in background patterns. The easiest way to switch is via a theme option:

```
\usetheme[pattern=triangle]{wildcat}
```

Section pages automatically use the same pattern as the title page. Available pattern options:

- **facet** — NU facets (default)
- **facetrand** — NU facets (alternate)
- **triangle** — triangle mosaic
- **triangleiso** — isometric triangles
- **trianglemesh** — wireframe mesh
- **squares** — square grid
- **circles** — circles
- **hexraise** — raised hex clusters
- **hexrand** — random hex shading
- **network** — network grid
- **quilt** — quilt rings
- **plain** — solid color

See **wildcat-gallery.pdf** for a visual preview of all patterns.

Colors

The theme has a few Northwestern-specific colors defined, which you can use in your slides. These are:

- nupurple
- nupurple90
- nupurple80
- nupurple70
- nupurple60
- nupurple50
- nupurple40
- nupurple30
- nupurple20
- nupurple10
- nupurple160
- nupurple150
- nupurple140
- nupurple130
- nupurple120
- nupurple110
- nurichblack
- nubrightgreen
- nubrightteal
- nubrightblue
- nubrightyellow
- nubrightorange
- nubrightred
- nudarkgreen
- nudarkteal
- nudarkblue
- nudarkyellow
- nudarkorange
- nudarkred

Modifying the Primary Color

Pass a `color` option to `\usetheme` — the theme auto-generates all lighter and darker shades automatically:

```
\usetheme[color=nudarkblue]{wildcat}
```

You can combine `color` and `pattern` in a single line:

```
\usetheme[color=nudarkblue, pattern=hexraise]{wildcat}
```

The `nu*` colors from the previous slide are all valid choices, as are the standard xcolor built-ins (`red`, `blue`, `cyan`, etc.). The alert and example colors (`wcalerted`, `wcexample`) are not affected and can be overridden separately.

Modifying the Primary Color: Custom RGB

The `color` option accepts any *named* color — not raw RGB values. If your brand color isn't already defined, declare it first:

```
\definecolor{myblue}{RGB}{0,53,107}
\usetheme[color=myblue]{wildcat}
```

The auto-generated shades follow a simple scale: `wcprimary10`–`40` mix the base color with white (lighter tints), and `wcprimary110`–`140` mix it with black (darker shades). These are sufficient for most use cases — if you need precise per-shade control, see the Advanced Customization section at the end of this document.

Using Colors in Graphs

Python Code

```
import matplotlib.pyplot as plt
import pandas as pd
plt.style.use('ggplot')
# Create colors
wcprimary = (78/255,42/255,132/255)
df = pd.read_stata("source/graphs/auto.dta")
plt.scatter(df["weight"], df["mpg"], color=wcprimary)
plt.xlabel("Weight")
plt.ylabel("MPG")
plt.savefig("source/graphs/plot-python.pdf")
plt.show()
```

Using Colors in Graphs

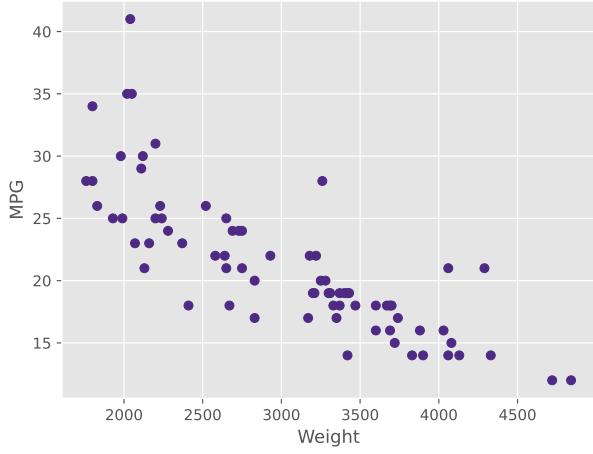


Figure: Python Graph Example

Using Colors in Graphs

R Code

```
library(ggplot2)
library(haven)
# Load data
df <- read_dta("source/graphs/auto.dta")
# Define custom RGB colors
wcprimary <- rgb(78/255, 42/255, 132/255)

# Plot mpg against weight, with marker color wcprimary
plot <- ggplot(df, aes(x = weight, y = mpg)) +
    geom_point(color = wcprimary) +
    labs(x = "Weight", y = "MPG")

# Save the plot to a file with a specific size
ggsave("source/graphs/plot-R.pdf", plot, width = 8, height = 5)
```

Using Colors in Graphs

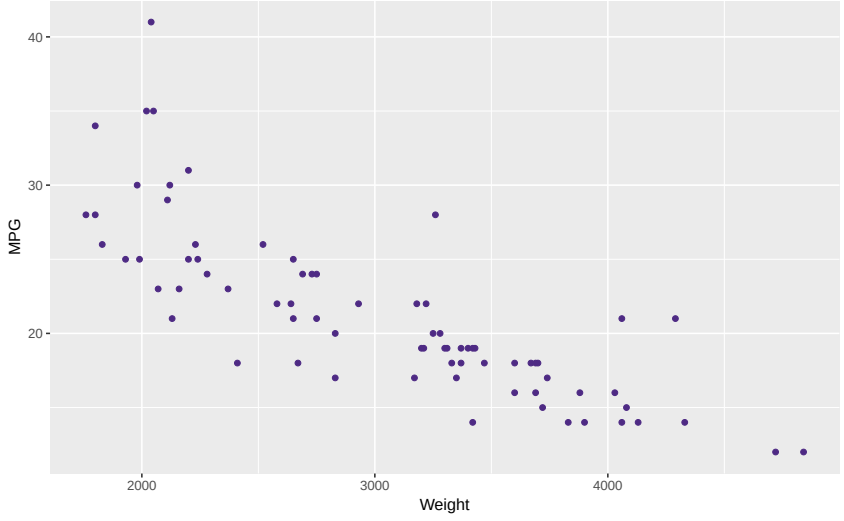


Figure: R Graph Example

Using Colors in Graphs

Stata Code

```
*ssc install schemepack, replace
sysuse auto, clear
local wcprimary "78 42 132"

twoway (scatter mpg weight, mcolor("`wcprimary'")), ///
        scheme(gg_tableau)

graph export "plot-stata.pdf", replace
```

Using Colors in Graphs

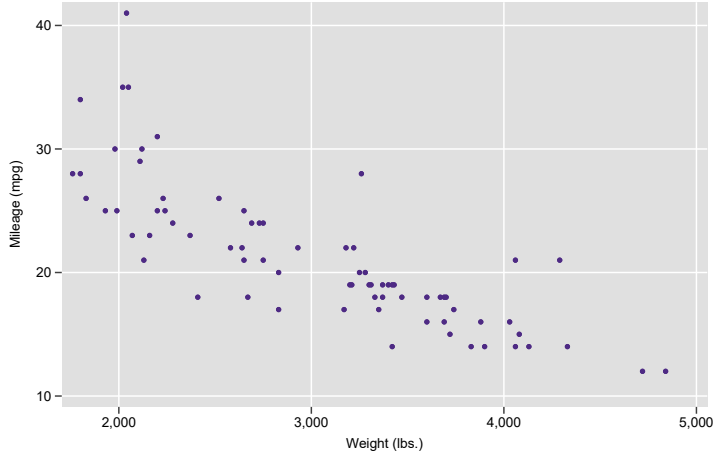


Figure: Stata Graph Example

Pattern Blocks (tcolorbox)

You can use tcolorbox style blocks with the background pattern instead of the default beamer blocks. You can create this with a `\begin{tblock}` environment. You can also use `talert` and `texample` for alert and example blocks, respectively.

T Block Title

This is a tcolorbox style block.

T Alert Title

This is a tcolorbox style alert block.

T Example Title

This is a tcolorbox style example block.

Custom Color Pattern Blocks

There is also a special block called `tpatternbox` which allows you to specify the color. This only works with non-primary (not red, green, or blue) colors, as you can't shade those easily.

Custom Pattern Block Title

This is a tpatternbox block. It can be created with the following code:

```
\begin{tpatternbox}[nudarkyellow]{Custom Pattern Block Title}
    This is a tpatternbox block...
\end{tpatternbox}
```

Box Examples (Default)

You can also just use the Beamer default blocks in the usual way. The default is non-rounded corners, non-shaded.

Main Block

This is an example block

Alert Box

This is an alert box

Example Box

This is an example box

Fonts

The theme uses Campton (titles) and Akkurat Pro (body) by default — these are proprietary fonts that must be purchased separately. Place the OTF files in a `fonts/` folder next to your `.tex` file:

```
your-presentation/
  your-slides.tex
  fonts/
    Campton/
      Campton Medium.otf
      Campton Light.otf
    Akkurat/
      AkkuratProLight-Regular.otf
      AkkuratProLight-Italic.otf
      AkkuratPro-Regular.otf
      AkkuratPro-Bold.otf
```

Requires **XeLaTeX** or **LuaLaTeX** — PDFLaTeX will fail.

Fonts: Overleaf / No Proprietary Fonts

If you are on Overleaf or do not have the proprietary fonts, use the `font=overleaf` option. This switches to TeX Gyre fonts, which are universally available with no files to upload:

```
\usetheme[font=overleaf]{wildcat}
```

You can combine all three options in one line:

```
\usetheme[font=overleaf, color=nudarkblue, pattern=triangle]{wildcat}
```

For advanced font customization (loading fonts manually, using system-installed fonts), see the Advanced Customization section at the end of this document.

Font Style: Examples

Here is what the body font will look like under normal usage:

- Regular
- *Italic*
- **Bold**
- ***Bold Italic***
- Alert
- *Alert Italic*
- Math:

$$e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n$$

Sections

You can use `\section{}` to group slides into sections. This will automatically add a section title slide at the beginning of each section.

You can turn this off by adding `\AtBeginSection{}` to your preamble (this tells beamer to do precisely nothing when a new section starts).

Advanced Customization

Advanced: Custom Fonts

If you have Campton and Akkurat Pro installed system-wide, use:

```
\usetheme[font=installed]{wildcat}
```

To load OTF files from a custom path (e.g. when uploading to Overleaf), add these lines to your preamble *after* `\usetheme`:

```
\newfontfamily\CamptonMedium[Path=fonts/Campton/, Extension=.otf]{Campton Medium}
\newfontfamily\CamptonLight[Path=fonts/Campton/, Extension=.otf]{Campton Light}
\setbeamerfont{title}{family=\CamptonMedium, size=\LARGE}
\setbeamerfont{author}{family=\CamptonLight, size=\small}
\setbeamerfont{frametitle}{family=\CamptonMedium, size=\Large}
\setbeamerfont{section}{family=\CamptonLight, size=\LARGE}
\setsansfont[Path=fonts/Akkurat/, UprightFont=*Light-Regular,
    ItalicFont=*Light-Italic, BoldFont=*-Regular,
    BoldItalicFont=*-Bold, Extension=.otf]{AkkuratPro}
```

Advanced: Fine-Grained Color Control

The auto-generated shades are a good starting point, but for precise brand colors you can override individual shades manually. Redefine any of the `wcprimary*` colors *after* `\usetheme`:

```
\usetheme{wildcat}                          % or with color= option
\definecolor{wcprimary}{RGB}{0,53,107}      % Yale Blue
\definecolor{wcprimary140}{RGB}{0, 34, 70}
\definecolor{wcprimary130}{RGB}{0, 40, 80}
\definecolor{wcprimary120}{RGB}{0, 45, 91}
\definecolor{wcprimary110}{RGB}{0, 50, 102}
\definecolor{wcprimary40}{RGB}{153, 174, 196}
\definecolor{wcprimary30}{RGB}{179, 194, 211}
\definecolor{wcprimary20}{RGB}{204, 215, 225}
\definecolor{wcprimary10}{RGB}{230, 235, 240}
```

Only the shades actually used by the theme need redefining: `wcprimary10`–`40` (block bodies) and `wcprimary110`–`140` (facet pattern segments).

Advanced: Custom Background Pattern (I)

To create your own pattern, define a command that draws TikZ commands filling the slide. The command should take one argument — the color — so that the theme can reuse it in different contexts. Here is a minimal example:

```
\newcommand{\mypattern}[1]{
    \fill[color=#1] (0,0) rectangle (\paperwidth,\paperheight);
}
```

The pattern command is called inside a **tikzpicture** environment whose bounding box already covers the full slide, so you only need to provide the drawing commands — no **\begin{tikzpicture}** wrapper is needed.

Any valid TikZ commands work. The built-in patterns each take one color argument and use **\the\paperwidth** / **\the\paperheight** to scale to the slide dimensions automatically.

Advanced: Custom Background Pattern (II)

Once your pattern command is defined, register it with the theme using two lines in your preamble *after* `\usetheme{wildcat}`:

```
\renewcommand{\bgpattern}{\mypattern{wcprimary}}
\renewcommand{\bgpatternblock}[1]{\mypattern{#1}}
```

Two commands are needed because the theme uses the pattern in two different ways:

- `\bgpattern` — title page, section slides, and standout frames. The color is fixed, so you bake it in when registering.
- `\bgpatternblock{color}` — pattern block headers (`tblock`, `talert`, `texample`, `tpatternbox`), where each block supplies its own color at render time.

Advanced: Custom Background Pattern (III)

The 12 built-in pattern definitions in `beamerinnerthemewildcat.sty` (lines 31–386) are a useful reference and starting point.

`\bgpatternplain` (line 384) is the simplest — a single `\fill` command — and is a good template to copy and modify. The other patterns show progressively more complex TikZ techniques: coordinate scaling, `\foreach` loops, random seeds, scopes, and drop shadows.

The `\bgpatternblock` variant used in `tblock` headers is typically identical to `\bgpattern`, just registered separately so the theme can pass the block's color through as an argument.

Here is a blank frame. You can use this to add a blank page to your presentation. You can specify this via:

```
\begin{frame}[plain]{}
...
\end{frame}
```

Standout Slides

You can use the `standout` command to create a slide with a large font and no title. This is useful for a slide that you want to stand out, but which doesn't need a title.

```
\standout{Questions?}
```

Questions?